

Habit Tracker with a Penalty System

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Abstract Most habit apps stop at reminders, and a reminder can be dismissed at no cost. This paper proposes a console-based habit tracker written in C++ that adds a penalty system. Users create time-based habits, which are tied to a fixed hour, and habit-based habits, which follow another habit within a grouped routine. The program records completed habits, shows reminders for habits that are due, and counts daily streaks. When a habit is skipped for a full day, the user must write a reflection of at least 100 words before using the rest of the program. The reflection covers why the habit was forgotten, what prevented it, and what the user will do to avoid forgetting it again. All data is stored in plain text files, and the program uses only basic C++ concepts: structs, arrays, loops, conditionals, and file input and output.

Index Terms C++, console application, habit tracker, penalty system, reflection, routine, streak.

I. INTRODUCTION

Most people know which habits are good for them. They know they should study daily, exercise, or sleep on time. The difficulty lies in doing these things consistently [1], [2]. University students face this problem more than most. After leaving the fixed routine of high school, their schedules change from day to day with classes, assignments, and organizational activities, and a habit with no fixed place in the day is easily forgotten.

Many habit apps exist, but most of them stop at a reminder. A reminder is easy to dismiss, and when nothing happens after a habit is skipped, skipping it again becomes easier. This project starts from a simple question: can an app make the user care about a habit and commit to it?

II. PROBLEM STATEMENT

This project addresses the following problems:

- 1) Habits are forgotten because they are not tied to a specific time or to a routine the user already has.
- 2) A plain checklist treats every habit separately, although habits usually occur together as a routine, such as waking up, drinking water, and then stretching.
- 3) Reminders carry no consequence, so the user can ignore them without any effect.
- 4) When a habit is missed, users rarely examine the reason, so the same mistake is repeated.

III. PROPOSED SOLUTION

This project proposes a simple console-based habit tracker written in C++. The program records every completed habit in a track record, shows reminders for habits that are due, and keeps a streak count of the consecutive days each habit has been done.

A. Habit Types and Activities

The user can add a habit in two ways. A time-based habit is linked to a fixed hour of the day. A habit-based habit is linked to another habit that must be completed before it [3]. Related habits are collected into a group called an activity, so that a routine can be followed in order. A flowchart of the add-habit process and of the overall program flow will be attached to the final submission.

B. Penalty System

The penalty system is the main feature of the program. When a habit is skipped for a full day, the user cannot use the rest of the program until they write a reflection of at least 100 words. The reflection must answer three questions: why the habit was forgotten, what prevented the user from doing it, and what concrete action the user will take to avoid forgetting it again [4]. After the reflection is saved, the streak of that habit is reset and the user can continue. This turns a missed habit into a point of review.

C. Data Storage and Scope

To keep the program simple, all data is saved in plain text files stored next to the program: one for habits, one for the track record, and one for reflections. The project will be built with basic C++ concepts, such as structs, arrays, loops, conditionals, and file input and output. Phone notifications are outside the scope of the project. Reminders appear inside the program when it starts and when the user opens the reminder menu.

IV. EXPECTED RESULT

The expected result is a working program that helps a user stay consistent with their habits. It combines reminders, streaks, grouped routines, and a reflection step that makes the user think before moving on.

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